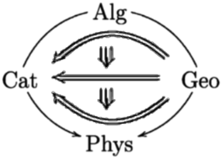
Algebraic Topology: Beyond Singular Cohomology

- Second Day-

Yuri Ximenes Martins[1]

Universidade Federal de Minas Gerais
Math-Phys-Cat Group



[1] webpage - nLab page

- Let us recall our plain...

Plain

1. General Ideas

Plain

1. General Ideas

Plain

1. General Ideas
   "*What is Algebraic Topology?*"

Plain

1. General Ideas
   "*What is Algebraic Topology?*"
   "*Why is Algebraic Topology the way is it?*"
2. Abstract Invariants

Plain

1. General Ideas
   "*What is Algebraic Topology?*"
   "*Why is Algebraic Topology the way is it?*"

2. Abstract Invariants

Plain

1. General Ideas
   "*What is Algebraic Topology?*"
   "*Why is Algebraic Topology the way is it?*"

2. Abstract Invariants
   "*What is Cohomology and Homotopy?*"

Plain

1. General Ideas
   "*What is Algebraic Topology?*"
   "*Why is Algebraic Topology the way is it?*"

2. Abstract Invariants
   "*What is Cohomology and Homotopy?*"
   "*Where are the Chain Complex Cohomologies?*"

Plain

1. General Ideas
   "*What is Algebraic Topology?*"
   "*Why is Algebraic Topology the way is it?*"
2. Abstract Invariants
   "*What is Cohomology and Homotopy?*"
   "*Where are the Chain Complex Cohomologies?*"
   "*Why not Homology?*"

Plain

1. General Ideas
   "*What is Algebraic Topology?*"
   "*Why is Algebraic Topology the way is it?*"
2. Abstract Invariants
   "*What is Cohomology and Homotopy?*"
   "*Where are the Chain Complex Cohomologies?*"
   "*Why not Homology?*"
3. Morphisms Between Abstract Invariants

Plain

1. General Ideas
   "*What is Algebraic Topology?*"
   "*Why is Algebraic Topology the way is it?*"
2. Abstract Invariants
   "*What is Cohomology and Homotopy?*"
   "*Where are the Chain Complex Cohomologies?*"
   "*Why not Homology?*"
3. Morphisms Between Abstract Invariants

Plain

1. General Ideas
   "*What is Algebraic Topology?*"
   "*Why is Algebraic Topology the way is it?*"

2. Abstract Invariants
   "*What is Cohomology and Homotopy?*"
   "*Where are the Chain Complex Cohomologies?*"
   "*Why not Homology?*"

3. Morphisms Between Abstract Invariants
   "*What are Characteristic Classes and Cohomology Operations?*"

Plain

1. General Ideas
   "*What is Algebraic Topology?*"
   "*Why is Algebraic Topology the way is it?*"

2. Abstract Invariants
   "*What is Cohomology and Homotopy?*"
   "*Where are the Chain Complex Cohomologies?*"
   "*Why not Homology?*"

3. Morphisms Between Abstract Invariants
   "*What are Characteristic Classes and Cohomology Operations?*"

4. Beyond the beyond

Reviewing

1. General Ideas

   "*What is Algebraic Topology?*"

   "*Why is Algebraic Topology the way is it?*"

Reviewing

1. General Ideas

   “What is Algebraic Topology?”

   “Why is Algebraic Topology the way is it?”

Reviewing

1. General Ideas

   "*What is Algebraic Topology?*"

> Algebraic Topology is the branch of Topology aiming to study based homotopy invariants defined by functors $F : \mathrm{Ho}(\mathbf{Top}_*) \to \mathbf{A}$,

"*Why is Algebraic Topology the way is it?*"

Reviewing

1. General Ideas

   “*What is Algebraic Topology?*”

Algebraic Topology is the branch of Topology aiming to study based homotopy invariants defined by functors $F : \mathrm{Ho}(\mathbf{Top}_*) \to \mathbf{A}$, specially hom-functors $[A, -]$ and $[-; A]$

“*Why is Algebraic Topology the way is it?*”

Reviewing

1. General Ideas

   "*What is Algebraic Topology?*"

> Algebraic Topology is the branch of Topology aiming to study based homotopy invariants defined by functors $F : \mathrm{Ho}(\mathbf{Top}_*) \to \mathbf{A}$, specially hom-functors $[A, -]$ and $[-; A]$ which are represented by H-monoids or H-comonoids.

   "*Why is Algebraic Topology the way is it?*"

Reviewing

1. General Ideas

   "*What is Algebraic Topology?*"

Algebraic Topology is the branch of Topology aiming to study based homotopy invariants defined by functors $F : \mathrm{Ho}(\mathbf{Top}_*) \to \mathbf{A}$, specially hom-functors $[A, -]$ and $[-; A]$ which are represented by H-monoids or H-comonoids.

"*Why is Algebraic Topology the way is it?*"

- the category of base spaces has null objects (instead of only initial and terminal objects);

Reviewing

1. General Ideas

   "*What is Algebraic Topology?*"

> Algebraic Topology is the branch of Topology aiming to study based homotopy invariants defined by functors $F : \mathrm{Ho}(\mathbf{Top}_*) \to \mathbf{A}$, specially hom-functors $[A, -]$ and $[-; A]$ which are represented by H-monoids or H-comonoids.

"*Why is Algebraic Topology the way is it?*"

- the category of base spaces has null objects (instead of only initial and terminal objects);
- the homotopy category has a weaker notion of isomorphism;

Reviewing

1. General Ideas

   "*What is Algebraic Topology?*"

Algebraic Topology is the branch of Topology aiming to study based homotopy invariants defined by functors $F : \mathrm{Ho}(\mathbf{Top}_*) \to \mathbf{A}$, specially hom-functors $[A, -]$ and $[-; A]$ which are represented by H-monoids or H-comonoids.

"*Why is Algebraic Topology the way is it?*"

- the category of base spaces has null objects (instead of only initial and terminal objects);
- the homotopy category has a weaker notion of isomorphism;
- hom-functors are very natural entities and they preserve homotopy limits

Reviewing

1. General Ideas

   "*What is Algebraic Topology?*"

Algebraic Topology is the branch of Topology aiming to study based homotopy invariants defined by functors $F : \mathrm{Ho}(\mathbf{Top}_*) \to \mathbf{A}$, specially hom-functors $[A, -]$ and $[-; A]$ which are represented by H-monoids or H-comonoids.

"*Why is Algebraic Topology the way is it?*"

- the category of base spaces has null objects (instead of only initial and terminal objects);
- the homotopy category has a weaker notion of isomorphism;
- hom-functors are very natural entities and they preserve homotopy limits

Reviewing

1. General Ideas

   "*What is Algebraic Topology?*"

Algebraic Topology is the branch of Topology aiming to study based homotopy invariants defined by functors $F : \mathrm{Ho}(\mathbf{Top}_*) \to \mathbf{A}$, specially hom-functors $[A, -]$ and $[-; A]$ which are represented by H-monoids or H-comonoids.

"*Why is Algebraic Topology the way is it?*"

- the category of base spaces has null objects (instead of only initial and terminal objects);
- the homotopy category has a weaker notion of isomorphism;
- hom-functors are very natural entities and they preserve homotopy limits, so that invariants defined by them are computable;
- hom-functors preserve H-monoids.

Reviewing

2. Abstract Invariants
   "*What is Cohomology?*"

Reviewing

2. Abstract Invariants
   "*What is Cohomology?*"

Reviewing

2. Abstract Invariants
   "*What is Cohomology?*"

In the abstract sence, the *cohomology with coefficients in* A is just $H(X;A) = [X;A]$.

Reviewing

2. Abstract Invariants
   "*What is Cohomology?*"

In the abstract sence, the *cohomology with coefficients in* A is just $H(X;A) = [X;A]$.

If A is n-deloopable, i.e., if there are $BA, B^2A, ..., B^nA$ such that $A \simeq \Omega BA \simeq \Omega^2 B^2 A \simeq ... \simeq \Omega^n B^n A$

Reviewing

2. Abstract Invariants
   "*What is Cohomology?*"

In the abstract sence, the *cohomology with coefficients in* A is just $H(X;A) = [X;A]$.

If A is n-deloopable, i.e., if there are $BA, B^2A, ..., B^nA$ such that $A \simeq \Omega BA \simeq \Omega^2 B^2 A \simeq ... \simeq \Omega^n B^n A$, then it is possible to define the higher A-valued cohomologies:

Reviewing

2. Abstract Invariants
   "*What is Cohomology?*"

In the abstract sence, the *cohomology with coefficients in* A is just $H(X; A) = [X; A]$.

If A is n-deloopable, i.e., if there are $BA, B^2A, ..., B^nA$ such that $A \simeq \Omega BA \simeq \Omega^2 B^2 A \simeq ... \simeq \Omega^n B^n A$, then it is possible to define the higher A-valued cohomologies:

$$H^i(X; A) := H(X; B^n A) = [X; B^n A]$$

Reviewing

2. Abstract Invariants
   "*What is Cohomology?*"

In the abstract sence, the *cohomology with coefficients in* A is just $H(X;A) = [X;A]$.

If A is n-deloopable, i.e., if there are $BA, B^2A, ..., B^nA$ such that $A \simeq \Omega BA \simeq \Omega^2 B^2 A \simeq ... \simeq \Omega^n B^n A$, then it is possible to define the higher A-valued cohomologies:

$$H^i(X;A) := H(X;B^nA) = [X;B^nA]$$

If A is infinitely deloopable we say that the corresponding cohomologies define the *A-valued cohomology theory* $H^*(X;A)$.

Reviewing

2. Abstract Invariants
   "*What is Homotopy of a Space?*"

Reviewing

2. Abstract Invariants
   "*What is Homotopy of a Space?*"

Reviewing

2. Abstract Invariants
   "*What is Homotopy of a Space?*"

Dually,

Reviewing

2. Abstract Invariants
   "*What is Homotopy of a Space?*"

Dually, the *abstract homotopy with coefficients in* A is just $\pi(X; A) = [A; X]$.

Dually, the *abstract homotopy with coefficients in* A is just $\pi(X;A) = [A;X]$.

Independently of A, the *higher A-valued homotopy groups* of X are defined by

$$\pi_n(X;A) := \pi(\Omega^n X;A) = [A;\Omega^n X]$$

Dually, the *abstract homotopy with coefficients in* A is just $\pi(X;A) = [A;X]$.

Independently of A, the *higher* A*-valued homotopy groups* of X are defined by

$$\pi_n(X;A) := \pi(\Omega^n X;A) = [A;\Omega^n X]$$

The loop space functor $\Omega : \mathbf{Top}_* \to \mathbf{Top}_*$ has a left-adjoint Σ, called *reduced suspension functor*.

Dually, the *abstract homotopy with coefficients in* A is just $\pi(X;A) = [A;X]$.

Independently of A, the *higher* A*-valued homotopy groups* of X are defined by

$$\pi_n(X;A) := \pi(\Omega^n X;A) = [A;\Omega^n X]$$

The loop space functor $\Omega : \mathbf{Top}_* \to \mathbf{Top}_*$ has a left-adjoint Σ, called *reduced suspension functor*. This means that for every based spaces X and Y we have

$$[\Sigma X;Y] \simeq [X;\Omega Y].$$

Dually, the *abstract homotopy with coefficients in A* is just $\pi(X;A)=[A;X]$.

Independently of A, the *higher A-valued homotopy groups* of X are defined by

$$\pi_n(X;A) := \pi(\Omega^n X;A) = [A;\Omega^n X]$$

The loop space functor $\Omega : \mathbf{Top}_* \to \mathbf{Top}_*$ has a left-adjoint Σ, called *reduced suspension functor*. This means that for every based spaces X and Y we have

$$[\Sigma X;Y] \simeq [X;\Omega Y].$$

In particular,

$$\pi_n(X;A) = [A;\Omega^n X] \simeq [\Sigma^n A;X] = \pi(X;\Sigma^n A).$$

Dually, the *abstract homotopy with coefficients in* A is just $\pi(X;A) = [A;X]$.

Independently of A, the *higher* A*-valued homotopy groups* of X are defined by

$$\pi_n(X;A) := \pi(\Omega^n X;A) = [A;\Omega^n X]$$

The loop space functor $\Omega : \mathbf{Top}_* \to \mathbf{Top}_*$ has a left-adjoint Σ, called *reduced suspension functor*. This means that for every based spaces X and Y we have

$$[\Sigma X;Y] \simeq [X;\Omega Y].$$

In particular,

$$\pi_n(X;A) = [A;\Omega^n X] \simeq [\Sigma^n A;X] = \pi(X;\Sigma^n A).$$

Reviewing

2. Abstract Invariants
   "*What is Homotopy of a Space?*"

Reviewing

2. Abstract Invariants
   "*What is Homotopy of a Space?*"

Reviewing

2. Abstract Invariants
   "*What is Homotopy of a Space?*"

Suspending the sphere we get another sphere: $\Sigma\mathbb{S}^n \simeq \mathbb{S}^{n+1}$,

Reviewing

2. Abstract Invariants
   "*What is Homotopy of a Space?*"

Suspending the sphere we get another sphere: $\Sigma\mathbb{S}^n \simeq \mathbb{S}^{n+1}$, so that

$$\pi_n(X) := \pi_n(X; \mathbb{S}^0) \simeq [\mathbb{S}^n; X].$$

Reviewing

2. Abstract Invariants
   "*What is Homotopy of a Space?*"

Suspending the sphere we get another sphere: $\Sigma\mathbb{S}^n \simeq \mathbb{S}^{n+1}$, so that

$$\pi_n(X) := \pi_n(X; \mathbb{S}^0) \simeq [\mathbb{S}^n; X].$$

This is what is typically called *nth-homotopy group of X*.

Reviewing

2. Abstract Invariants
   "*What is Homotopy of a Space?*"

Suspending the sphere we get another sphere: $\Sigma\mathbb{S}^n \simeq \mathbb{S}^{n+1}$, so that

$$\pi_n(X) := \pi_n(X; \mathbb{S}^0) \simeq [\mathbb{S}^n; X].$$

This is what is typically called *nth-homotopy group of* X.

Notice that $\mathbb{S}^1 \simeq I/\partial I$, where $I = [0, 1]$.

Reviewing

2. Abstract Invariants
   "*What is Homotopy of a Space?*"

Suspending the sphere we get another sphere: $\Sigma\mathbb{S}^n \simeq \mathbb{S}^{n+1}$, so that

$$\pi_n(X) := \pi_n(X; \mathbb{S}^0) \simeq [\mathbb{S}^n; X].$$

This is what is typically called *nth-homotopy group of X*.

Notice that $\mathbb{S}^1 \simeq I/\partial I$, where $I = [0, 1]$.

Furthermore, for every space (X, X_0) with a distinguished subspace $X_0 \subset X$ and every based space Y we have

$$[X/X_0; Y] \simeq [(X, X_0); Y].$$

Reviewing

2. Abstract Invariants
   "*What is Homotopy of a Space?*"

Suspending the sphere we get another sphere: $\Sigma\mathbb{S}^n \simeq \mathbb{S}^{n+1}$, so that

$$\pi_n(X) := \pi_n(X; \mathbb{S}^0) \simeq [\mathbb{S}^n; X].$$

This is what is typically called *nth-homotopy group of X*.

Notice that $\mathbb{S}^1 \simeq I/\partial I$, where $I = [0,1]$.

Furthermore, for every space (X, X_0) with a distinguished subspace $X_0 \subset X$ and every based space Y we have

$$[X/X_0; Y] \simeq [(X, X_0); Y].$$

Thus, $\pi_1(X)$ is the homotopy classes of closed paths in X

Reviewing

2. Abstract Invariants
   "*What is Homotopy of a Space?*"

Suspending the sphere we get another sphere: $\Sigma\mathbb{S}^n \simeq \mathbb{S}^{n+1}$, so that

$$\pi_n(X) := \pi_n(X; \mathbb{S}^0) \simeq [\mathbb{S}^n; X].$$

This is what is typically called *nth-homotopy group of X*.

Notice that $\mathbb{S}^1 \simeq I/\partial I$, where $I = [0, 1]$.

Furthermore, for every space (X, X_0) with a distinguished subspace $X_0 \subset X$ and every based space Y we have

$$[X/X_0; Y] \simeq [(X, X_0); Y].$$

Thus, $\pi_1(X)$ is the homotopy classes of closed paths in X, i.e., it is the fundamental group of X.

Remark.

Remark. All of the above makes sense in the context of Abstract Homotopy Theory,

Remark. All of the above makes sense in the context of Abstract Homotopy Theory, i.e., if one replace **Top** with some other category **C** complete and cocomplete category with null-objects and a "model structure",

Reviewing

Remark. All of the above makes sense in the context of Abstract Homotopy Theory, i.e., if one replace **Top** with some other category **C** complete and cocomplete category with null-objects and a "model structure", meaning that it has some coherent notion of homotopy between morphisms, leading us to consider its homotopy category.

Remark. All of the above makes sense in the context of Abstract Homotopy Theory, i.e., if one replace **Top** with some other category **C** complete and cocomplete category with null-objects and a “model structure”, meaning that it has some coherent notion of homotopy between morphisms, leading us to consider its homotopy category.

Not only this:

Reviewing

Remark. All of the above makes sense in the context of Abstract Homotopy Theory, i.e., if one replace **Top** with some other category **C** complete and cocomplete category with null-objects and a "model structure", meaning that it has some coherent notion of homotopy between morphisms, leading us to consider its homotopy category.

Not only this: *every general discussion that will be presented next extends to the context of Abstract Homotopy Theory.*

Abstract Invariants

Where is Singular Cohomology?

Theorem.

Lemma[Whitehead Lemma]. *The homotopy type of a CW-complex id determined by its homotopy groups.*

Theorem.

1. *Abelian topological groups G are ∞-loop spaces*

Theorem.

1. *Abelian topological groups G are ∞-loop spaces*
2. *The G-valued cohomology theory*

$$H^*(X;G) = [X; B^nG]$$

is just singular cohomology with coefficients in G.

Theorem.

1. *Abelian topological groups* G *are* ∞*-loop spaces*
2. *The* G*-valued cohomology theory*

$$H^*(X;G) = [X;B^nG]$$

is just singular cohomology with coefficients in G*.*

Theorem.

1. *Abelian topological groups G are ∞-loop spaces*
2. *The G-valued cohomology theory*

$$H^*(X;G) = [X;B^nG]$$

is just singular cohomology with coefficients in G.

Lemma[Whitehead Lemma]. *The homotopy type of a CW-complex id determined by its homotopy groups.*

Theorem.

1. *Abelian topological groups G are ∞-loop spaces*
2. *The G-valued cohomology theory*

$$H^*(X;G) = [X; B^n G]$$

is just singular cohomology with coefficients in G.

Lemma[Whitehead Lemma]. *The homotopy type of a CW-complex id determined by its homotopy groups.*

- Thus, when we give a list of homotopy groups we are fixing a CW complex up to homotopy.

proof of Theorem.

proof of Theorem. From Whitehead lemma, for every abelian group G and every $n \geq 0$ one can define a CW-complex $K(n;G)$ by

$$\pi_i(K(n;G)) = \begin{cases} G, & \text{if } i = n \\ 0, & \text{otherwise.} \end{cases}.$$

proof of Theorem. From Whitehead lemma, for every abelian group G and every $n \geq 0$ one can define a CW-complex $K(n;G)$ by

$$\pi_i(K(n;G)) = \begin{cases} G, & \text{if } i = n \\ 0, & \text{otherwise.} \end{cases}.$$

They are called *Eilenberg-Maclane spaces of* G.

proof of Theorem. From Whitehead lemma, for every abelian group G and every $n \geq 0$ one can define a CW-complex $K(n;G)$ by

$$\pi_i(K(n;G)) = \begin{cases} G, & \text{if } i = n \\ 0, & \text{otherwise.} \end{cases}.$$

They are called *Eilenberg-Maclane spaces of* G. "**OBS**: *The abelian hypothesis on* G *is needed because* $\pi_i(X)$ *is abelian for every* X *if* $i > 1$".

proof of Theorem. From Whitehead lemma, for every abelian group G and every $n \geq 0$ one can define a CW-complex $K(n; G)$ by

$$\pi_i(K(n; G)) = \begin{cases} G, & \text{if } i = n \\ 0, & \text{otherwise.} \end{cases}.$$

They are called *Eilenberg-Maclane spaces of G*. "**OBS**: *The abelian hypothesis on G is needed because* $\pi_i(X)$ *is abelian for every X if* $i > 1$". Notice that

proof of Theorem. From Whitehead lemma, for every abelian group G and every $n \geq 0$ one can define a CW-complex $K(n;G)$ by

$$\pi_i(K(n;G)) = \begin{cases} G, & \text{if } i = n \\ 0, & \text{otherwise.} \end{cases}.$$

They are called *Eilenberg-Maclane spaces of* G. "**OBS**: *The abelian hypothesis on* G *is needed because* $\pi_i(X)$ *is abelian for every* X *if* $i > 1$". Notice that

$$\begin{aligned} \pi_i(\Omega K(n+1;G)) &= [\mathbb{S}^i; \Omega K(n+1;G)] \simeq [\Sigma \mathbb{S}^i; K(n+1;G)] \\ &\simeq [\mathbb{S}^{i+1}; K(n+1;G)] = \pi_{i+1}(K(n+1;G)) \\ &\simeq \pi_i(K(n;G)). \end{aligned}$$

proof of Theorem. Thus, each $K(n;G)$ is deloopable, with

$$BK(n;G) = K(n+1;G).$$

proof of Theorem. Thus, each $K(n;G)$ is deloopable, with

$$BK(n;G) = K(n+1;G).$$

Since G is a topological group, $K(0;G) \simeq G$ as a topological space.

proof of Theorem. Thus, each $K(n;G)$ is deloopable, with

$$BK(n;G) = K(n+1;G).$$

Since G is a topological group, $K(0;G) \simeq G$ as a topological space. Therefore, $G \simeq \Omega^n K(n;G)$ for every n, i.e.,

proof of Theorem. Thus, each $K(n;G)$ is deloopable, with

$$BK(n;G) = K(n+1;G).$$

Since G is a topological group, $K(0;G) \simeq G$ as a topological space. Therefore, $G \simeq \Omega^n K(n;G)$ for every n, i.e., it is a ∞-loop space.

Thanks